

EE6320 RF Integrated Circuits

Project: Mixer Design

Mixer Performance Summary Table

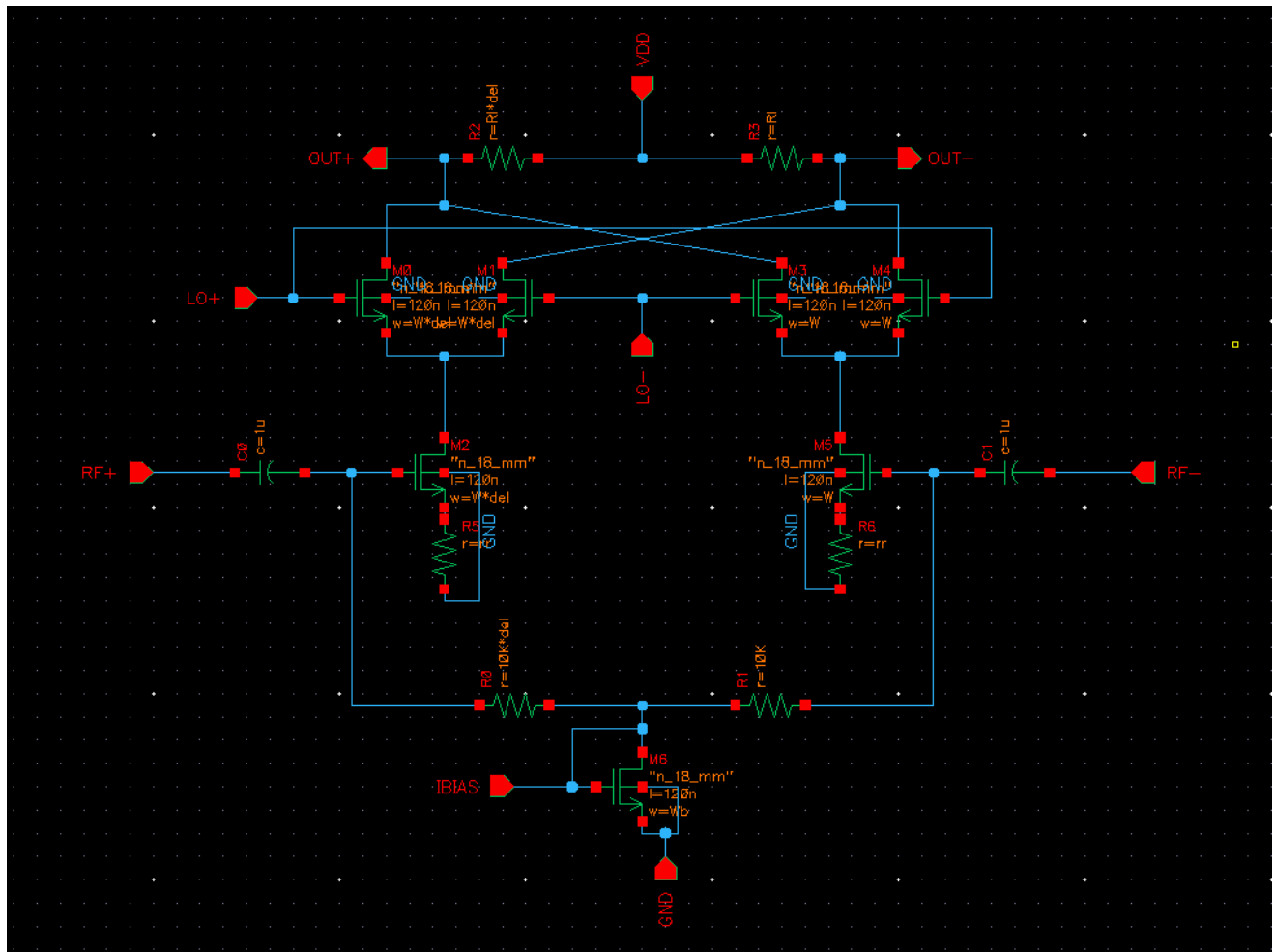
	Design Metric	Performance	Specification
Conversion Gain	Gain in the specified band [$f_{RF} = f_{LO}$]	20.772 dB	>20 dB
	Gain in the specified band [$f_{RF} = f_{LO} + 10\text{MHz}$]	20.77 dB	>20 dB
	Peak Gain flatness in specified band [Max-Min Gain]	0.002 dB	-
	3dB RF Bandwidth [From the plot of $f_{RF} = f_{LO}$]	1.5516 GHz	-
Noise Figure	Integrated SSB Noise Figure for $f_{LO} = 3.5\text{ GHz}$	11.5069	$\leq 12\text{ dB}$
	RF Frequency Band over which SSB NF<10dB $f_{LO}=3.5\text{GHz}$	5.35 MHz	$\leq 12\text{ dB}$
Linearity - IIP2	Input power used for extrapolation	-40 dBm	-
	Power of Fundamental Tone at output (at chosen input power)	-19.22129 dBm	-
	Power of IM2 Tone at output (at chosen input power)	-92.3378 dBm	-
	Extrapolated IIP2	33.1435 dBm	$\geq +30\text{dBm}$
Linearity - IIP3	Input power used for extrapolation	-40 dBm	-
	Power of Fundamental Tone at output (at chosen input power)	-19.0717 dBm	-
	Power of IM3 Tone at output (at chosen input power)	-89.8838 dBm	-
	Extrapolated IIP3	-4.41532 dBm	$\geq 0\text{ dBm}$
Power (I+Q)	Mixer DC power consumption [Excluding Bias]	593 μW	Minimize
	Bias circuit power consumption	4.8 mW	Minimize
Other	Sum of all resistances [excluding bias]	11.4 K Ω	-
	Sum of biasing resistances	40 k Ω	-
	Sum of all capacitances [Including AC coupling]	4 μF	-
	Sum of all inductances	0	
	Load chosen (each R_load)	2.7 K Ω	-
	Differential Mixer Input Capacitance (C_gs Caps)	60 fF	-
	Simulator Used	Spectre	-

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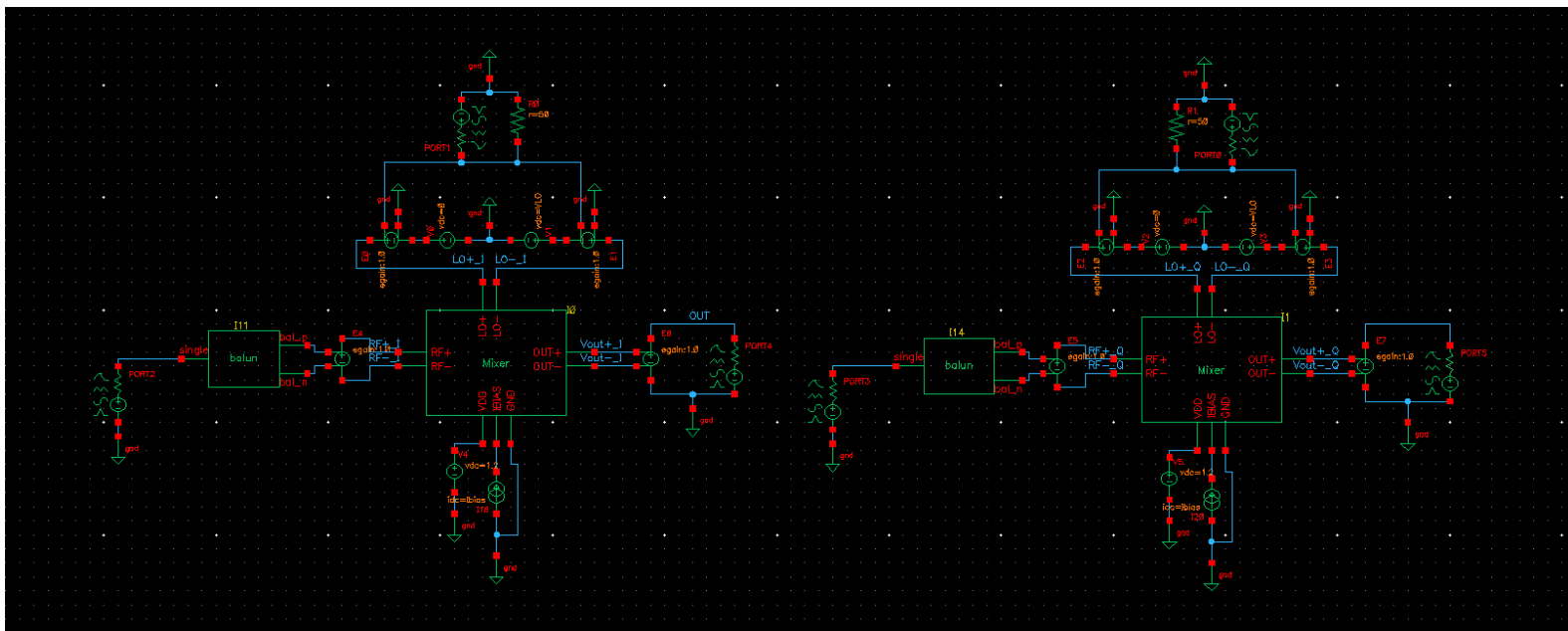
LNA + Mixer Performance Summary Table

	Design Metric	LNA	Mixer	Cascade	
				Expected	Simulated
Conversion Gain	$f_{IN} = f_{LO}, f_{LO} = 3.5 \text{ GHz}$	36.56 dB	20.772 dB	57.332 dB	46.157 dB
	$f_{IN} = f_{LO} + 10\text{MHz}, f_{LO} = 3.5 \text{ GHz}$	36.09 dB	20.77 dB	56.86 dB	46.064 dB
	$f_{IN} = f_{LO} - 10\text{MHz}, f_{LO} = 3.5 \text{ GHz}$	34.58 dB	20.771 dB	55.351 dB	46.2 dB
Noise Figure	$f_{IN} = f_{LO}, f_{LO} = 3.5 \text{ GHz}$	1.80 dB	9.77 dB	1.78 dB	1.52 dB
	$f_{IN} = f_{LO} + 10\text{MHz}, f_{LO} = 3.5 \text{ GHz}$	1.80 dB	10 dB	1.8 dB	1.53 dB
	$f_{IN} = f_{LO} - 10\text{MHz}, f_{LO} = 3.5 \text{ GHz}$	1.80 dB	10 dB	1.8 dB	1.53 dB
Linearity IIP3	Input power used for extrapolation	-40 dBm	-40 dBm	-	-40 dBm
	Power of Fundamental Tone at output (at chosen input power)	-3.36 dBm	-19.0717 dBm	-	
	Power of IM_3 Tone at output (at chosen input power)	-49.19 dBm	-89.8838 dBm	-	
	Extrapolated IIP_3	-9.523 dBm	-4.41532 dBm	-	
Power	Total power consumption [Excluding Bias]	2.456 mW	593 μ W	3.049 mW	3.061788 mW
	Bias circuit power consumption	2.4 mW	2.4 mW	4.8 mW	4.82 mW

Mixer Schematic



Mixer Testbench



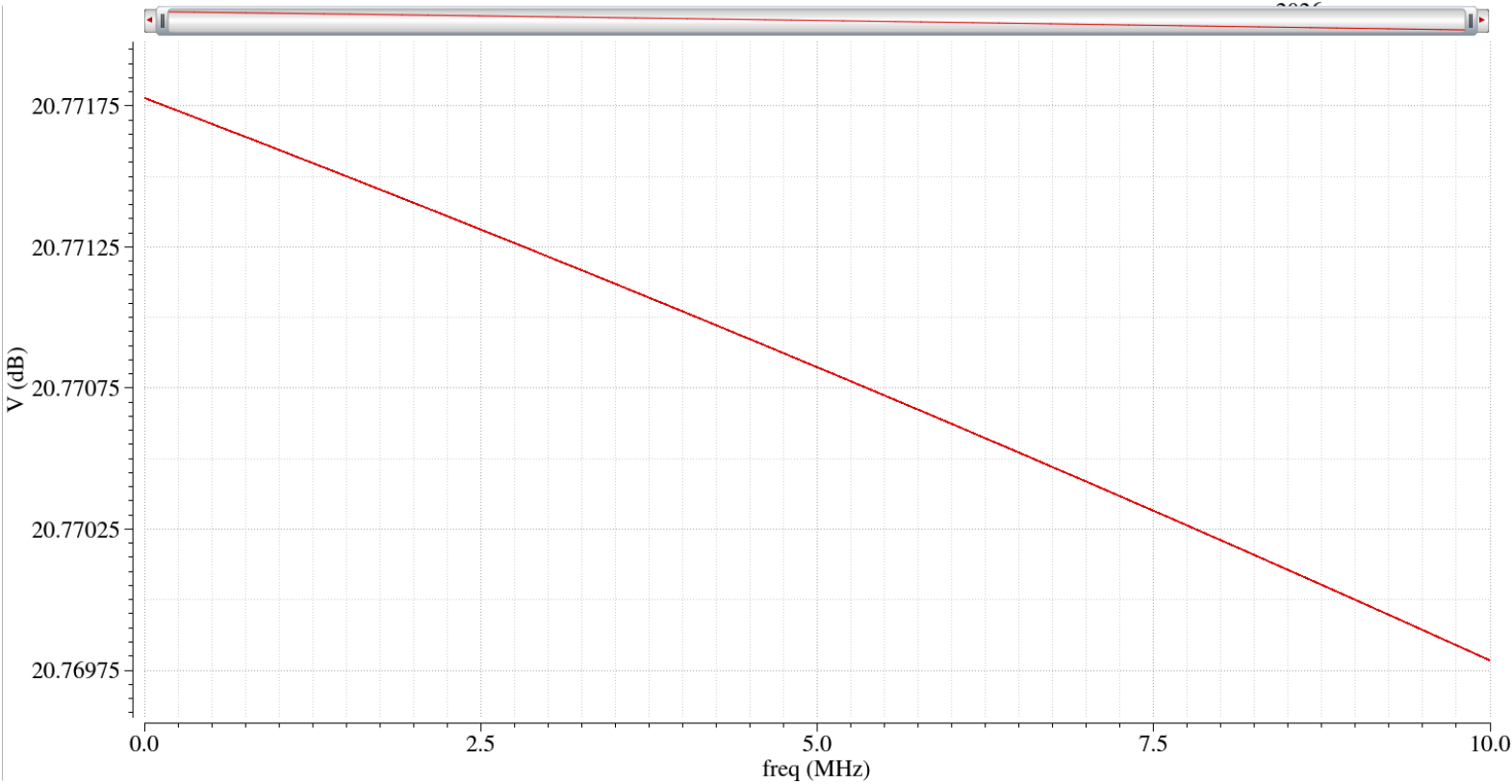
Design Variable Values

Design Variable	Value
Resistance (load, each side)	2.7 kΩ
Length of all MOS	120 nm
Width of Switch MOS (all 4)	48 μm
Width of Transconductance MOS (both MOS)	48 μm
Bias Current	2 mA
V_LO Amplitude	700 mV

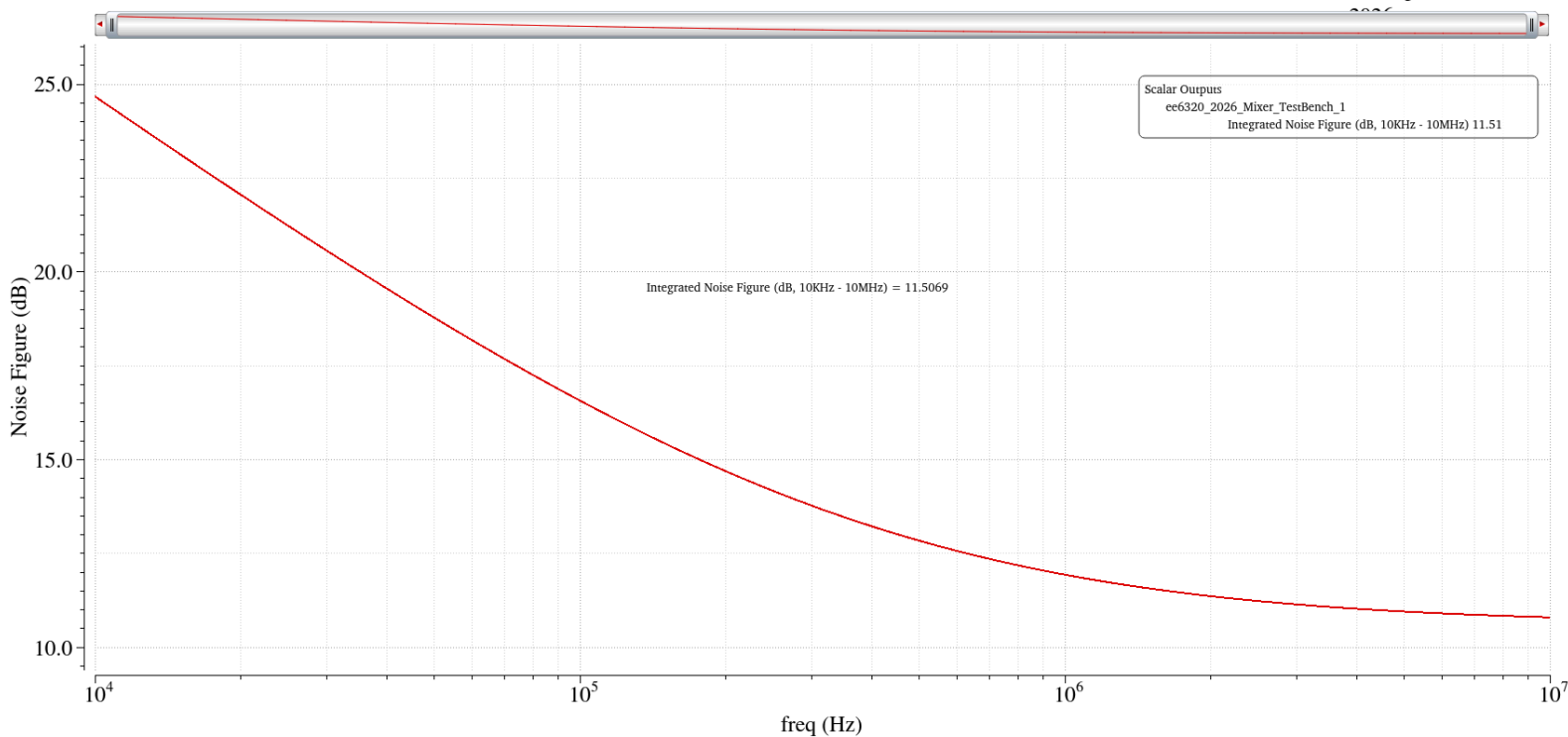
Fixed Constant Parameters

• Transconductance MOS C _{gs} : 60 fF
• Current Mirror MOS: W = 96 μm, L = 120 nm
• Coupling capacitances: 1 μF (each)
• Bias Resistances: 10 KΩ (each)

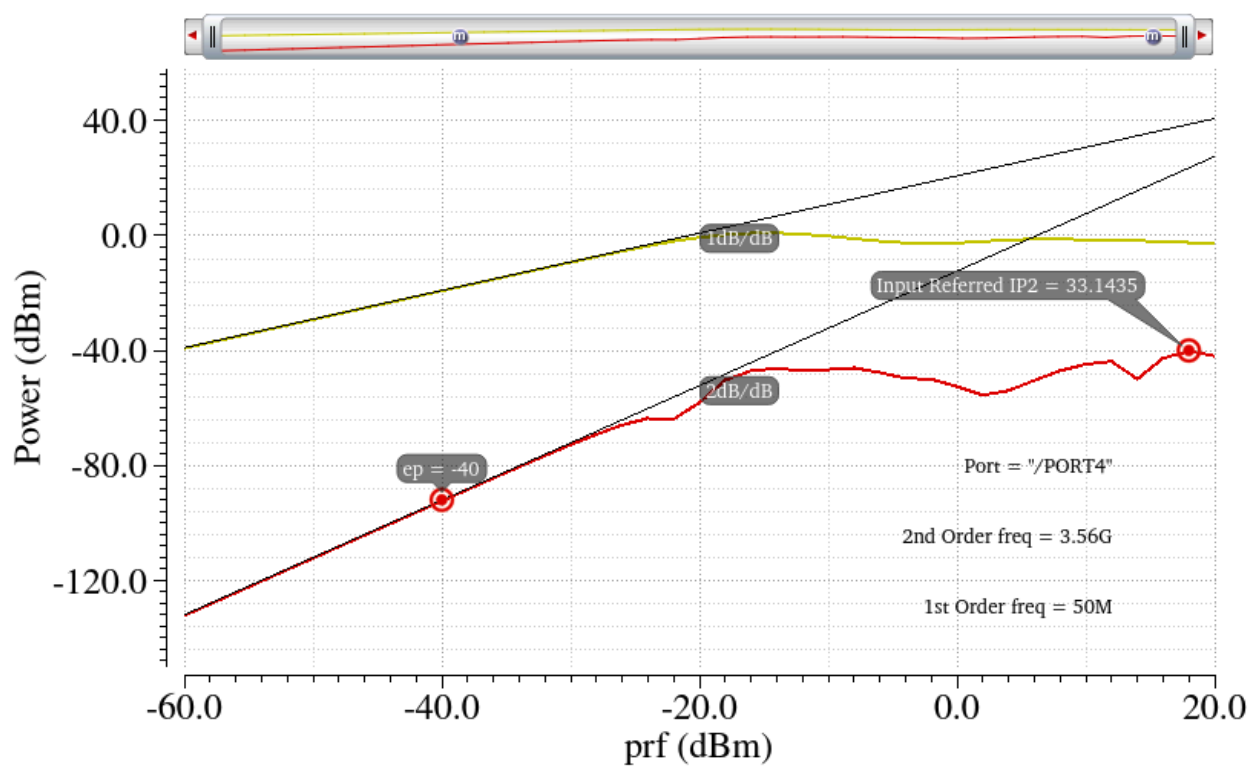
Conversion Gain of Mixer at 3.5 GHz



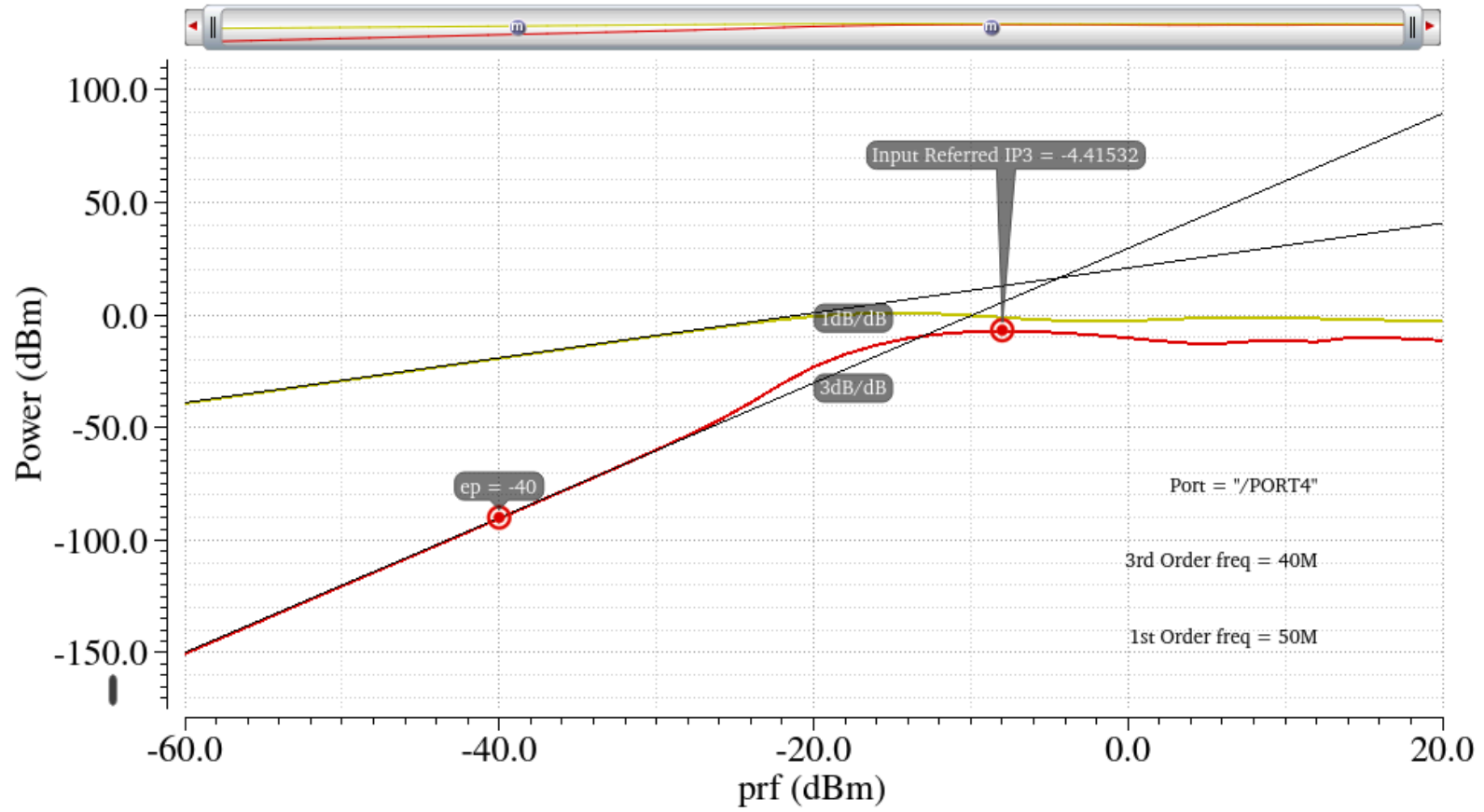
Noise Figure of Mixer at 3.5 GHz



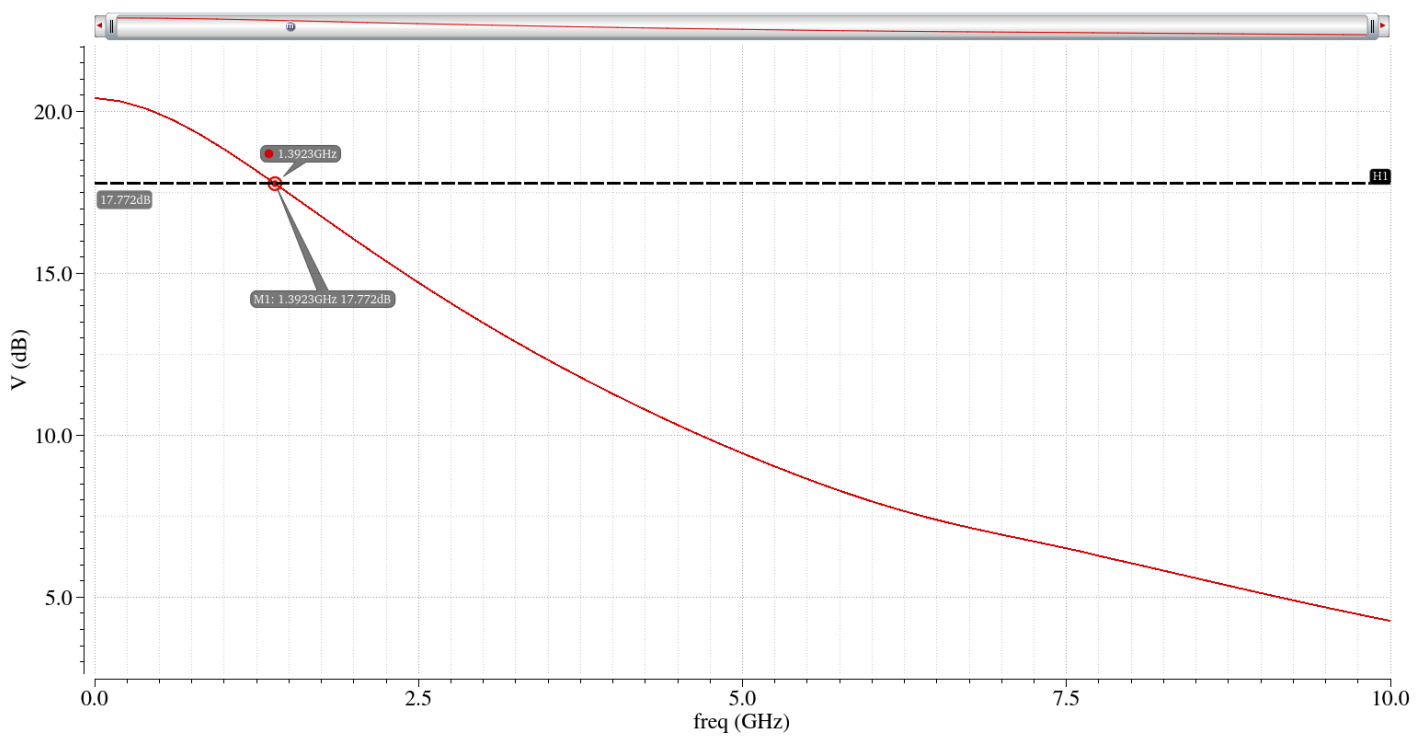
Mixer IIP2 Linearity



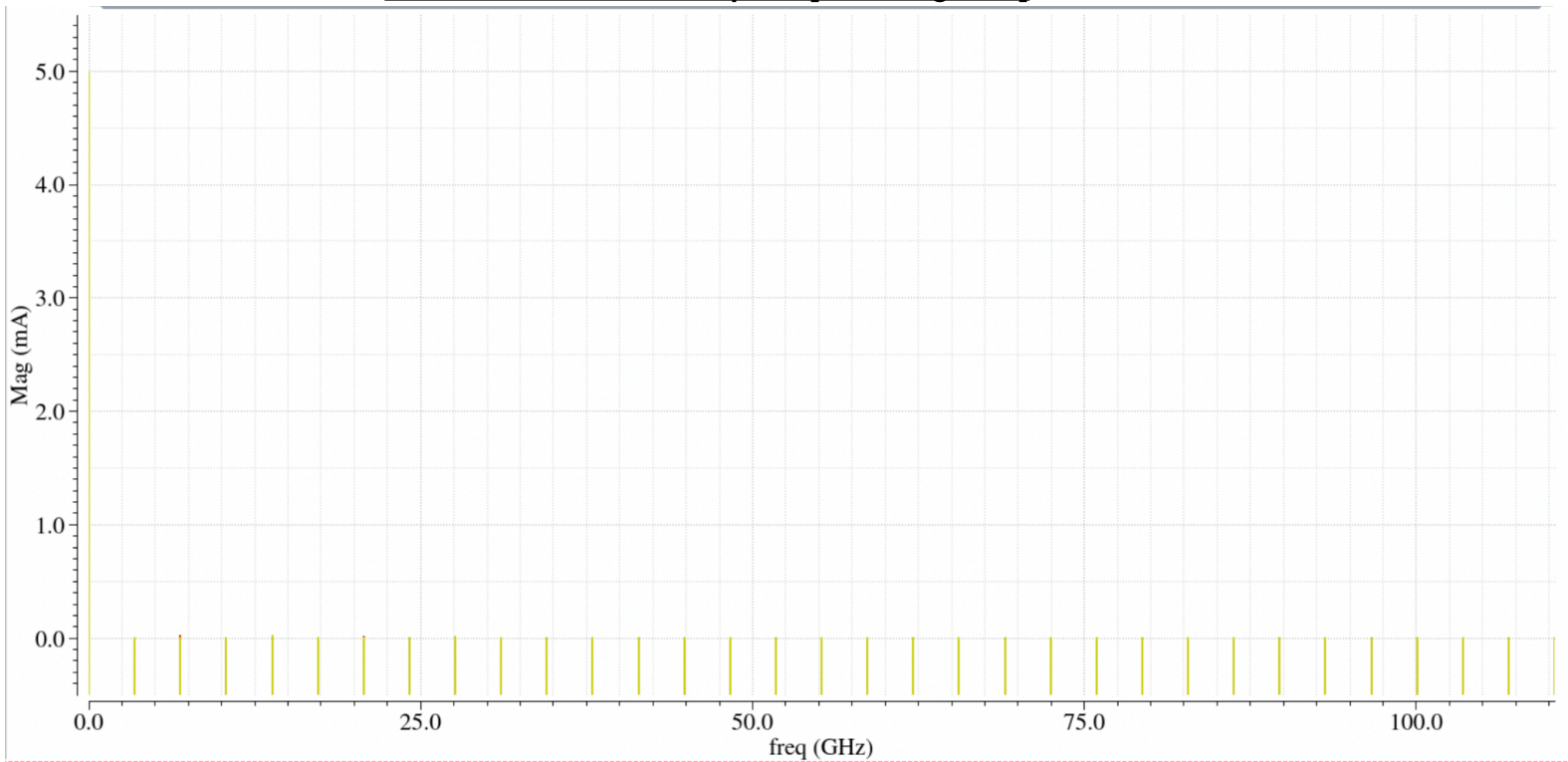
Mixer IIP3 Linearity



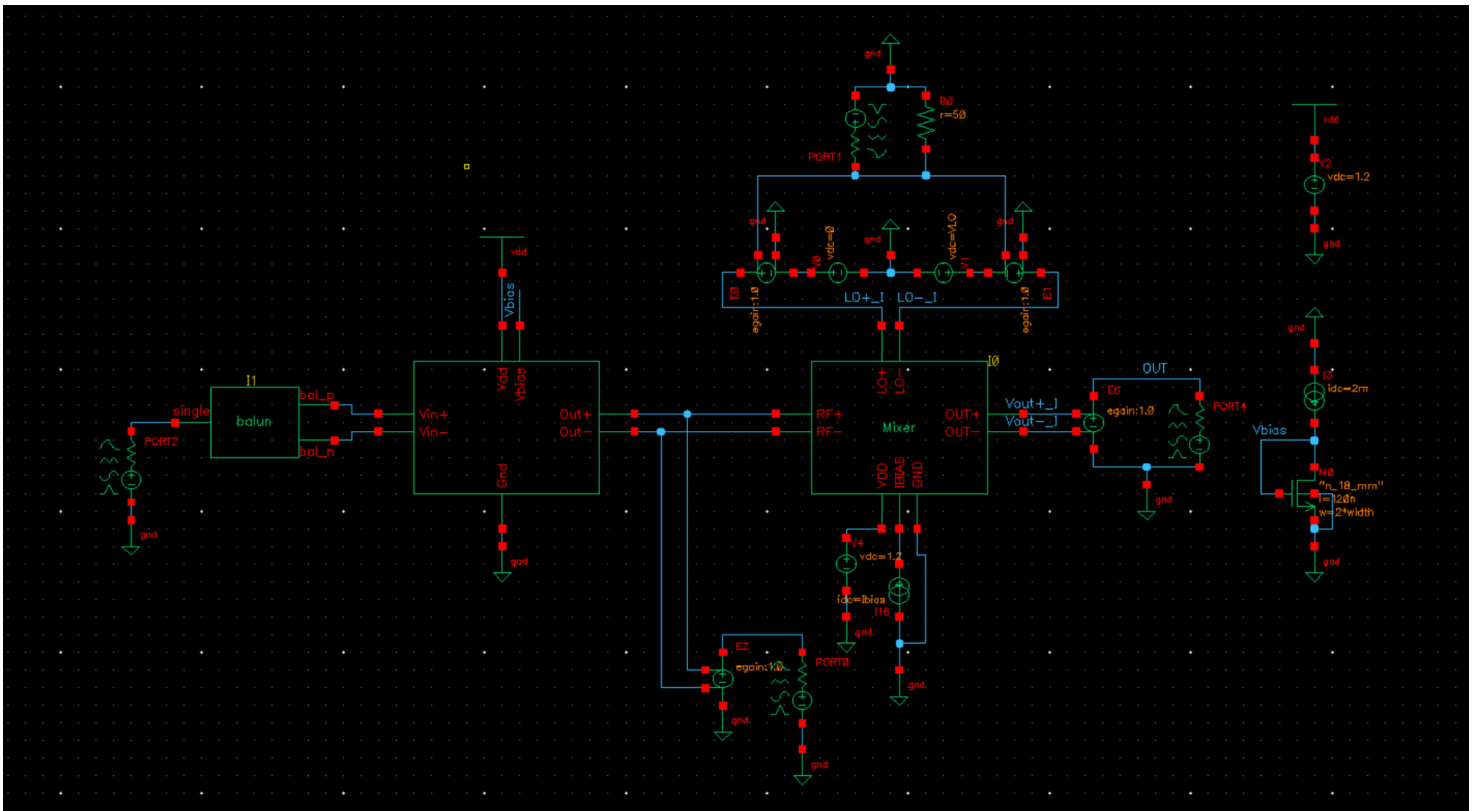
Mixer 3-dB Bandwidth



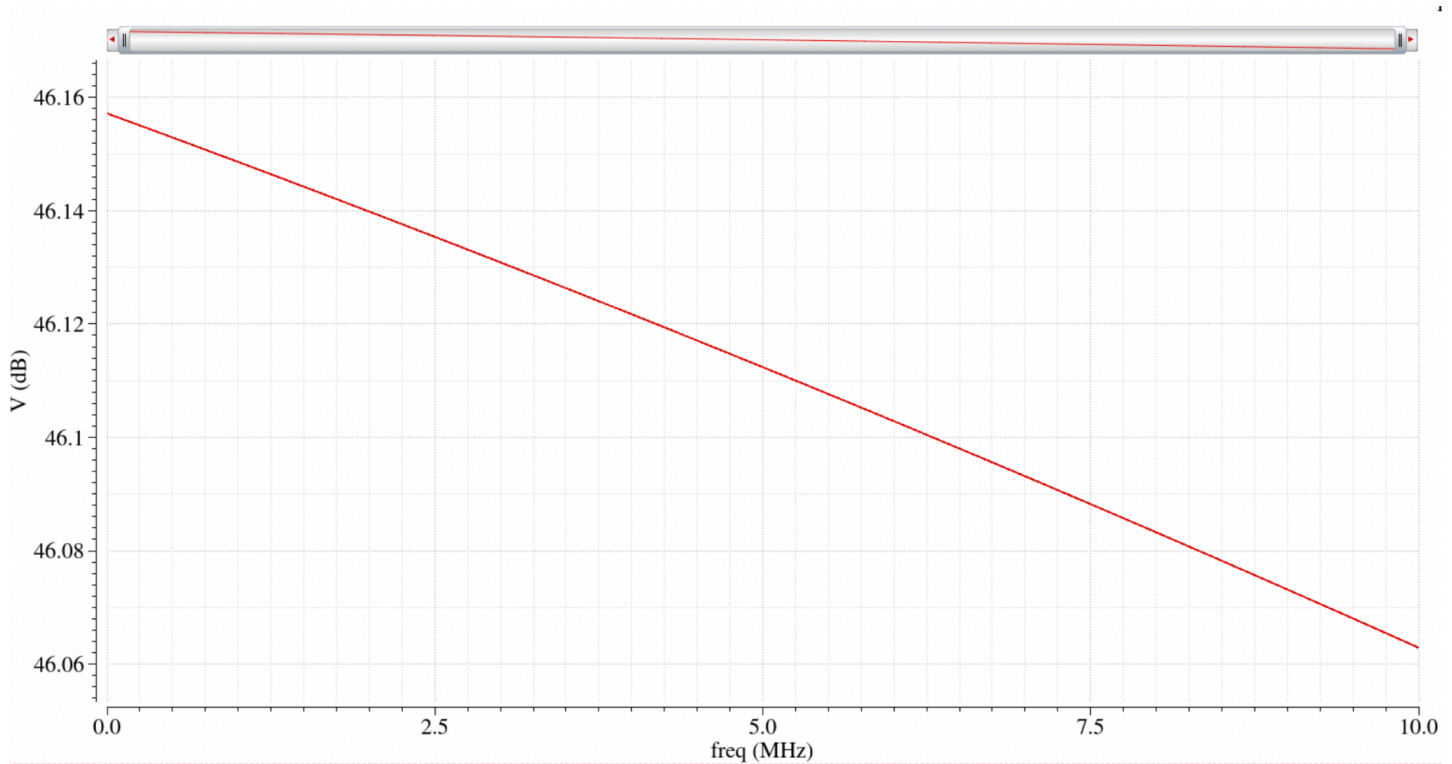
Mixer DC Power Consumption [Including Bias]



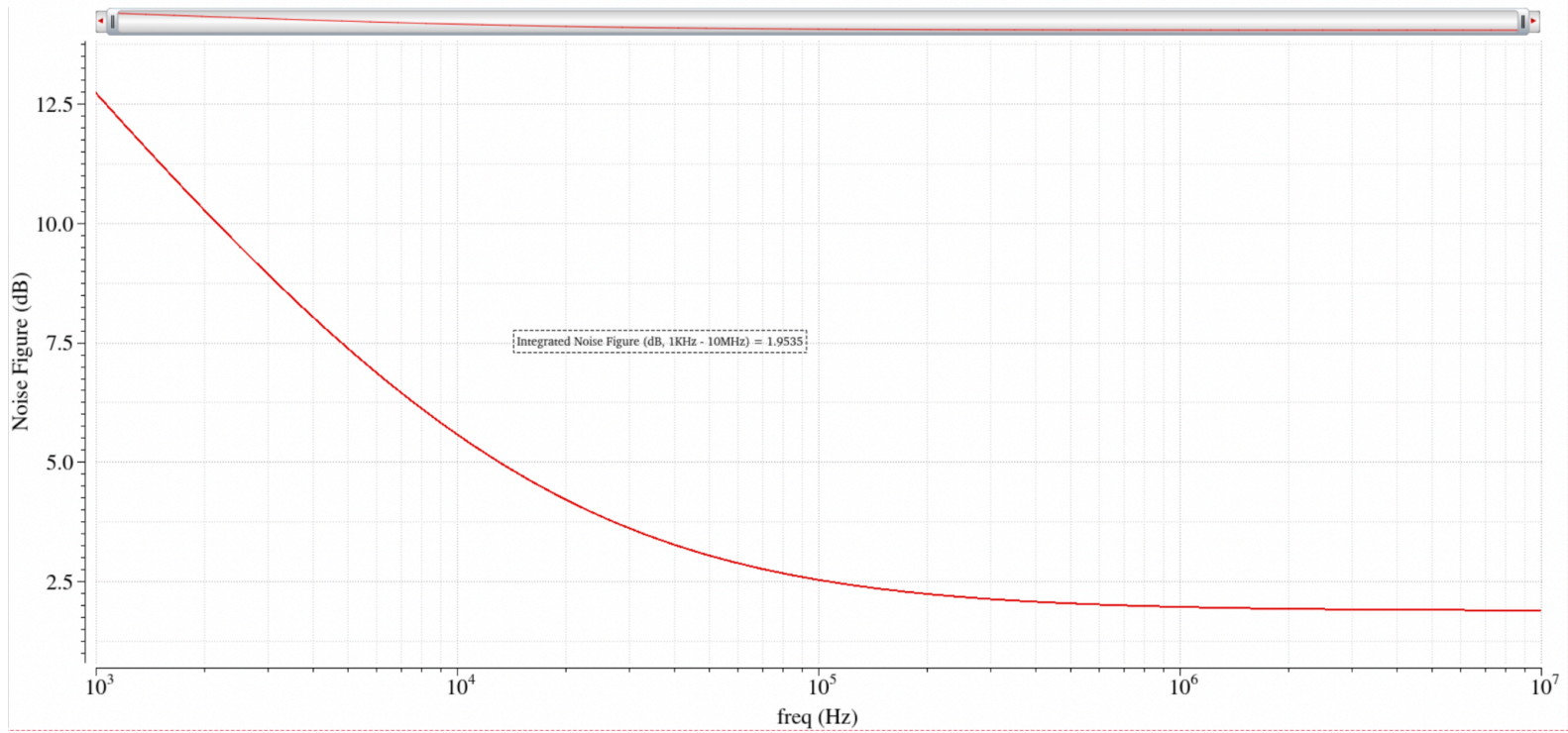
LNA + Mixer Testbench



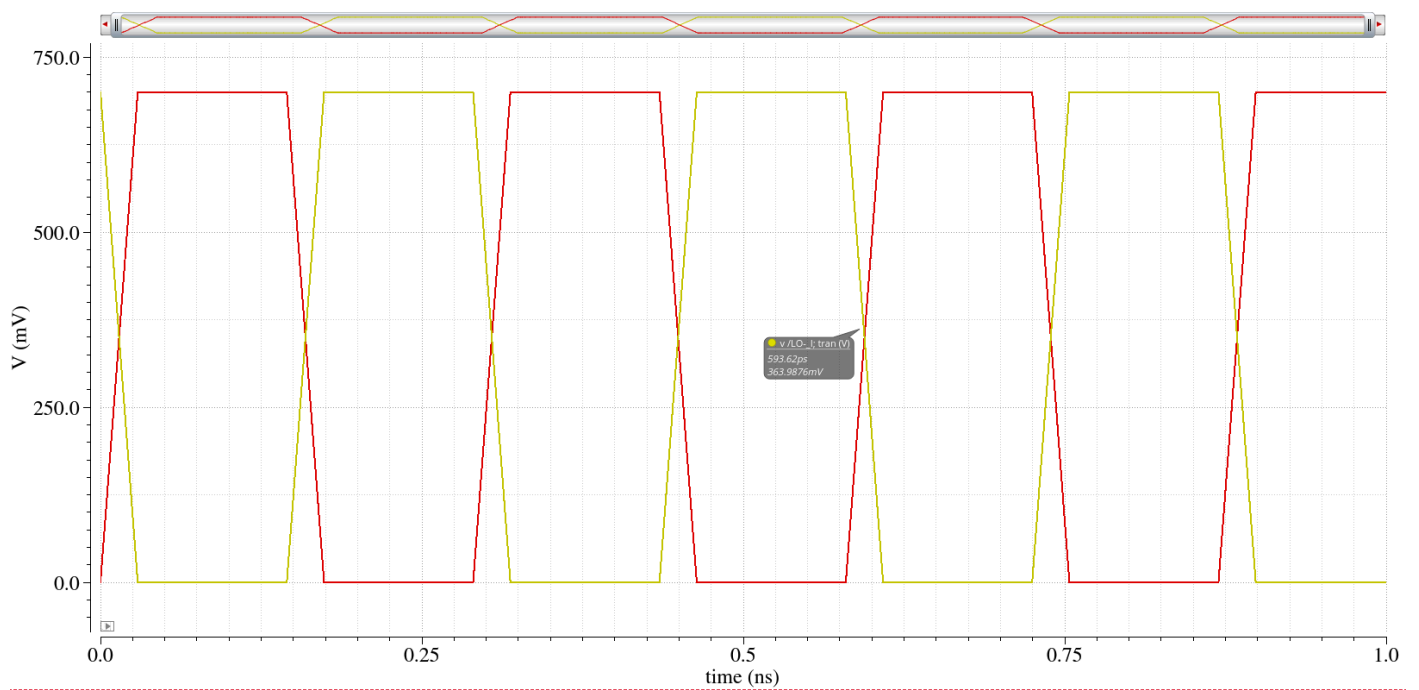
LNA + Mixer IIP3 Linearity
Simulation took forever to run
LNA + Mixer Conversion Gain at 3.5 GHz



LNA + Mixer Noise Figure at 3.5 GHz



LO Waveform Characteristics



LO Characteristics taken are (for generalized manner, as f_{lo} varies for various analysis):

- Frequency: 3.5 Hz
- Time period: 285.71 ps
- Delay time: 0 s (for I Lo signal), 71.42 ps (for Q Lo signal)
- Pulse Width: 114.28 ps
- Rise/Fall Time: 28.57 ps

Calculations done for the Project

mixer all calculations.

req:- IIP2 :- ≥ 20dBm

IIP3 :- ≥ -5dBm

iter:- I_{bias} = 62.5 μA R_s = 50 Ω → IIP₂ = 13.94
 67.5 μA :- R_s = 50 Ω → IIP₃ = 17.6

R_D = 150 → 18.247dB → source degeneration

Gain:- $\frac{2}{\pi} g_m R \rightarrow \frac{2}{\pi} \times 2.7 \text{ K} \Omega \times g_m = 10$

$g_m = 5.817 \text{ mS}$

$$\frac{2}{\pi}$$

$$V_i = \sqrt{\frac{2}{\pi}} \sqrt{R} \sqrt{V_i - V_{th}} \sqrt{g_m}$$

IIP₂

-6.716

-7.269

$$V_o = R(V_i - V_{th})g_m$$

$$V_{th} = (V_i - V_{th})g_m R$$

$$V_{th} = V_i \left(\frac{g_m R}{1 + g_m R} \right)$$

$$V_o = -R \left(\frac{1}{1 + g_m R} \right) g_m$$

with source degeneration

Noise Figure:-

$$F_{dB} \text{ eqn:- } F_{total} = F_1 + \frac{F_2 - F_1}{G_1}$$

$$= 1.51 + \frac{9.5 - 1}{4530} = 1.5119 \quad @ 3.56 \text{ Hz}$$

$$N.F = 1.78 \text{ dB}$$

$$N.F = 1.8 \text{ dB}$$

$$N.F = 1.8 \text{ dB}$$

$$@ 3.66 \text{ Hz}$$

$$@ 3.44 \text{ Hz}$$

Procedure adopted for the Project

- *At starting I assumed RF mosfet and switch mosfets widths as 50u. and run the dc analysis and calculated the gm value of rf mosfet and calculated load resistance. Almost my gain and NF reached the requirements.*
- *I got more required current so I decreased the rf mosfet width to 20u and increased switch mosfet width to get better VLO swing limits. After takking these value my iip3 got worst so I tried changing my VLO and checked the VLO swing limits.*
- *So I made gm3 of rf mosfet to 0 by using calculator by sweeping vgs. Then based on that I changed my current. And by changing the VLO i got iip3 < 5dBm.*
- *For minimizing power consumption I changed the bias current mosfet width by 10 times of rf width and changed the current by 10.*

Path to the Project Files

- **Path for Mixer Schem -**
/home/ns25z205/cadence/EE6320_2026/ee6320_2026/Mixer_V1
- **Path for Mixer TestBench -**
/home/ns25z205/cadence/EE6320_2026/ee6320_2026/Mixer_TestBench
- **LNA + MIXER test bench -**
/home/ns25z205/cadence/EE6320_2026/ee6320_2026/LNA#2bMixer